

Benjamin Abensur

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EDUCATION

EPFL — Swiss Federal Institute of Technology in Lausanne

Expected Graduation: 2028

Bachelor of Science in Communication Systems

- Relevant Coursework: Introduction to Machine Learning (5.75/6), Algorithms I (5.5/6), Linear Algebra (5.5/6), Probability & Statistics, Numerical Methods for Visual Computing & ML, Computer Architecture, Computer Systems, Responsible Software

WORK EXPERIENCE

Biosynex, Incoming AI Engineer (part-time contractor); Remote (Euronext-listed rapid diagnostics)

Starting Oct 2026

- Engaged to introduce the executive team to applied AI, then build AI tools into internal workflows.

EPFL, Teaching Assistant, Mechanics (PHYS-101) and Linear Algebra (MATH-111)

Sep 2026 – Present

- Selected on academic merit for a Mechanics course of **1,000+** students (CS, Chemistry, EE); lead a weekly 2-hour exercise session and answer questions on the course's Ed forum.

BABOO, Founder & Event Producer; Lausanne, Switzerland

2025

- Produced **3** club nights end to end; one night at Noche Club (May 2025) brought in **CHF 5,000** with an internationally touring DJ.

NanoSynex (Technion spin-off), AI & Automation (project-based), reporting to the CEO; Remote

2024 (3 months)

- Deployed two offline LLMs (Ollama; Mistral 7B, Llama 3 8B) on the CEO's Macs behind a router with a shared Obsidian memory, so confidential R&D and investor documents never left the machine (GDPR).
- Built Python scripts triaging the CEO's Gmail and Outlook inboxes with an LLM classifier, before native AI email integrations existed.
- Harmonised and cleaned the test reader's experiment exports (Python, pandas), then matched them against a partner veterinary lab's reference results to measure agreement rates for a validation study.

Sneaker Resale Venture, Partner (team of 4)

2019 – 2021

- Secured a license to Flare AIO, a sell-out EU sneaker bot, by automating its restock purchase; ran it on Foot Locker and Snipes drops.
- Resold the license for about twice its cost; generated roughly **€10k** over the period.

PROJECTS

Gaming Addiction Prediction — ML from scratch, CS-233 (team of 3)

github.com/Benabens/addiction-classifier-numpy

Python, NumPy, no ML libraries

- Predicted gaming-addiction level (3 classes) and score (0–10) from gaming and mental-health data with KNN, logistic/linear regression, K-Means and an MLP with hand-written backpropagation; two written reports.
- Inverse-frequency weighted cross-entropy lifted rarest-class recall from **0 to 0.69** and macro-F1 from **0.55 to 0.76** (5-fold CV).

ReCHor — public-transport journey planner, CS-108

github.com/Benabens/ReCHor

Java, JavaFX

- Finds every Pareto-optimal journey between two Swiss stops (Connection Scan Algorithm) over real CFF timetables (~258 MB).
- Memory-mapped, bit-packed timetable storage; JavaFX UI with autocomplete and iCalendar export; 44 classes, ~3,600 lines of tests.

ICoop — two-player cooperative 2D game, CS-107

github.com/Benabens/ICoop

Java

- Fire-and-water co-op game with a boss fight; interactions via double dispatch (visitor pattern). ~3,700 lines.

Kairo — AI career CRM

kairo-internships.vercel.app

Next.js, TypeScript, Python, PostgreSQL (Supabase, row-level security), Claude API, GitHub Actions, Vercel

- Designed and shipped a daily pipeline aggregating **16,000+** internships, jobs and programmes from **35** career boards, Indeed/LinkedIn (30 markets) and hackathon feeds, each scored 0–100 by rules + an LLM.
- LLM extraction of deadlines and eligibility, each answer backed by a verbatim quote; 160+ tests, CI, deployed on Vercel.
- Contact CRM turning freeform notes into structured contacts via Claude, then linking each job to the people I know there and prompting who to follow up with.

Cortex — AI exam-preparation platform

cortex-exam.vercel.app · GitHub

Next.js, TypeScript, PostgreSQL, Drizzle, Claude API, LaTeX, Railway

- Learns each course's exam format from past papers, weights every topic by how heavily it was examined, schedules spaced revision on weak points, and generates faithful practice exams via a multi-pass LLM pipeline; used it to prepare my own exams in **3** EPFL courses.
- Answers accepted only if verified by sandboxed code execution or symbolic checks; per-user Postgres schemas, cost caps; 330+ tests.

Portfolio — personal website (Next.js, TypeScript, Vercel)

benjamin-abensur.vercel.app

- Designed and built my personal site end to end: projects, background and afro-house productions, every figure traceable to a repo.

Other Projects: Verilog digital-logic library (42/42 self-checking testbenches); networking and UNIX-filesystem project in C (CS-202, in a pair); VeriReason, a PyTorch Transformer from first principles (in progress); Reed-Solomon QR-code generator (Java, CS-107).

LEADERSHIP & COMMUNITY INVOLVEMENT

Scout Summer Camp, Co-Camp Director; Mévouillon, France

Jul 2025

- Co-directed a one-month camp for **100** children from two local groups (Nice, Strasbourg); led a team of **20** counsellors, promoted team leads and wrote the camp's full educational project.
- Co-managed the **€80k** camp budget, running the €18k food line myself (local farmers, Intermarché); built the site at pre-camp: tents, dry toilets, wooden tables; organised outings and explorations.

EPFL Linear Algebra Bootcamp (MATH-111), Co-instructor; Lausanne, Switzerland

Nov 2025

- Gave the lectures and wrote the exercise sets of a paid one-week bootcamp for **30** first-year students (mid-semester break).

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, Verilog, x86/RISC-V assembly

ML & AI: PyTorch, NumPy, pandas, matplotlib, Claude API, Ollama

Web & Systems: Next.js, React, Node.js, REST APIs, PostgreSQL, Supabase, web scraping, GitHub Actions, Linux, POSIX sockets, Git

Spoken: French (native), English (fluent), Hebrew (basic), Spanish (basic)